

Tutorial 7: FINA2205

Portfolio Theory and Asset Pricing 3

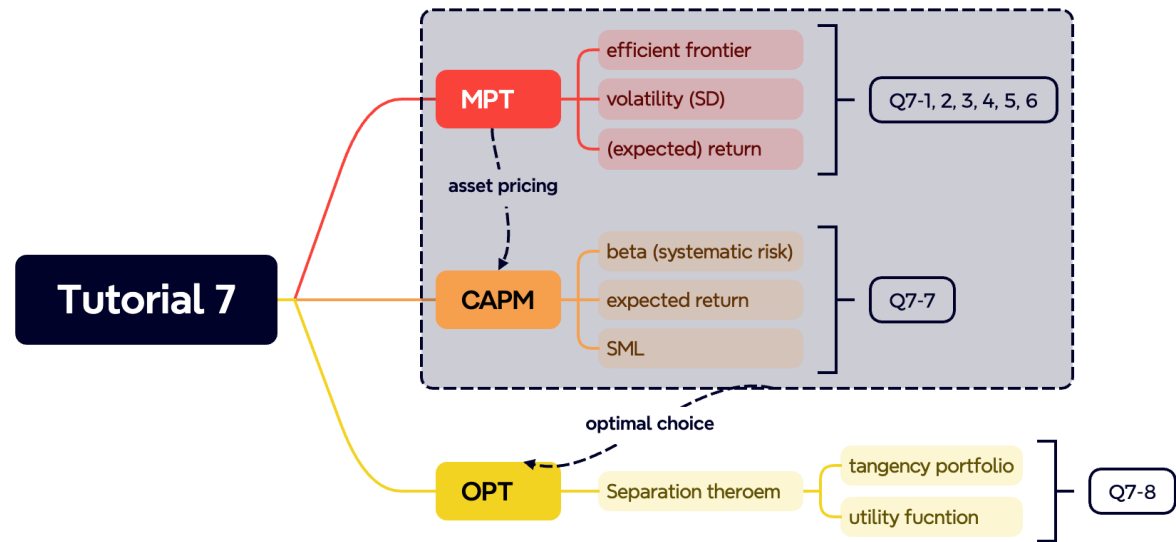
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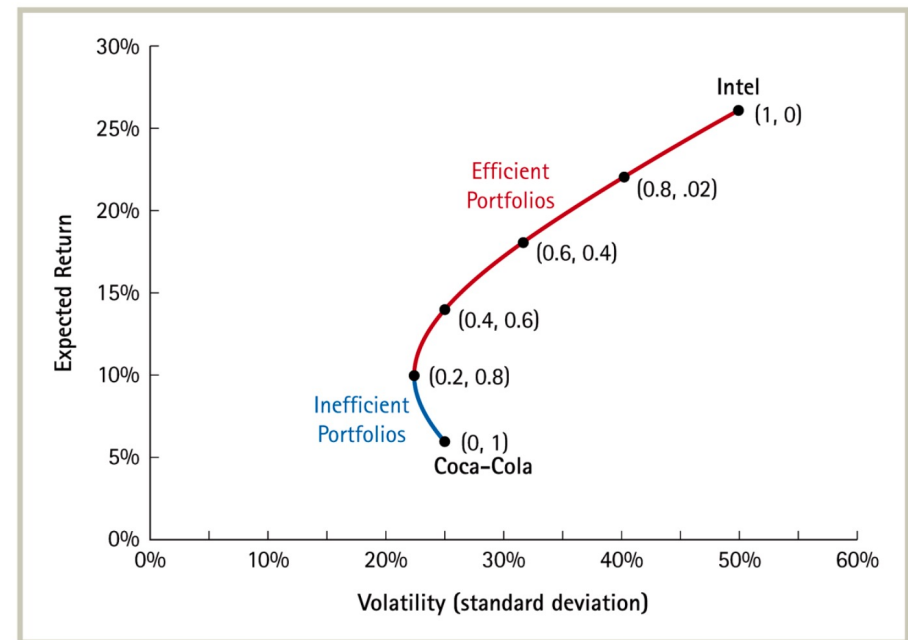
Agenda

- 8 problems



Recap: Modern portfolio theory

- Goal: Combining risky assets to optimize the risk-return trade-off through **diversification**.
- The feasible set: By constantly changing the portfolio weights, we map out all possible risk-return combinations.
- The efficient frontier: The optimal upper segment (the red curve).



Recap: Modern portfolio theory

Core formulas (**two-asset risky portfolio**):

- Expected return (linear):

$$\mu_p = k\mu_A + (1 - k)\mu_B$$

- Portfolio volatility (non-linear):

$$\sigma_p = \sqrt{k^2\sigma_A^2 + 2k(1 - k)\sigma_A\sigma_B\rho_{AB} + (1 - k)^2\sigma_B^2}$$

- Key takeaway: The correlation coefficient (ρ_{AB}) is the entire engine of diversification.

TABLE 11.4

Expected Returns and Volatility for Different Portfolios of Two Stocks

Portfolio Weights		Expected Return (%)	Volatility (%)
x_I	x_C	$E[R_p]$	$SD[R_p]$
1.00	0.00	26.0	50.0
0.80	0.20	22.0	40.3
0.60	0.40	18.0	31.6
0.40	0.60	14.0	25.0
0.20	0.80	10.0	22.3
0.00	1.00	6.0	25.0

Caveat: Typically, in a standard long-only portfolio, a lower correlation (ρ_{AB}) enhances diversification benefits and reduces portfolio risk. However, this relationship flips during **short selling**. When you have a short position, lowering the ρ_{AB} will actually increase portfolio risk!

7-2

a standard long-only portfolio

2. Arbor Systems and Gencore stocks both have a volatility of 50%. Compute the volatility of a portfolio with 50% invested in each stock if the correlation between the stocks is (a) +1.0, (b) 0.50, (c) 0, (d) -0.50, and (e) -1.0. In which of the cases is the volatility lower than that of the original stocks?

Review Only

Click the icon to see the Worked Out Solution (Formula Solution).

If the correlation is +1.0, the volatility of the portfolio is _____%. (Round to one decimal place.)

If the correlation is 0.50, the volatility of the portfolio is _____%. (Round to one decimal place.)

If the correlation is 0, the volatility of the portfolio is _____%. (Round to one decimal place.)

If the correlation is -0.50, the volatility of the portfolio is _____%. (Round to one decimal place.)

If the correlation is -1.0, the volatility of the portfolio is _____%. (Round to one decimal place.)

In which of the cases is the volatility lower than that of the original stocks? (Select the best choice below.)

- ☐ A. In cases (d) and (e).
- ☐ B. In all of the cases.
- ☐ C. In cases (b), (c), (d) and (e).
- ☐ D. In none of the cases.

7-2 solution

Setup:

- $k = 0.5, 1 - k = 0.5$
- $\sigma_A = 50\%, \sigma_B = 50\%$

$$\sigma_p = \sqrt{k^2 \sigma_A^2 + 2k(1 - k)\sigma_A \sigma_B \rho_{AB} + (1 - k)^2 \sigma_B^2}$$

- Different Scenarios ($\rho_{AB} = +1, +0.25, 0, -0.5, -1$)

7-2 solution

Scenario 1: $\rho = +1$

Calculation:

$$\sigma_P = \sqrt{0.5^2 50^2 + 2 \times 0.5 \times 0.5 \times 50 \times 50 \times 1 + 0.5^2 50^2} = 50\%$$

$$\sigma_P = \sigma_A = \sigma_B$$

Scenario 2: $\rho = +0.5$

Calculation:

$$\sigma_P = \sqrt{0.5^2 50^2 + 2 \times 0.5 \times 0.5 \times 50 \times 50 \times 0.5 + 0.5^2 50^2} = 43.30\%$$

$$\sigma_P < \sigma_A = \sigma_B$$

7-2 solution

Scenario 3: $\rho = 0$

Calculation:

$$\sigma_P = \sqrt{0.5^2 50^2 + 2 \times 0.5 \times 0.5 \times 50 \times 50 \times 0 + 0.5^2 50^2} = 35.36\%$$

$$\sigma_P < \sigma_A = \sigma_B$$

Scenario 4: $\rho = -0.5$

Calculation:

$$\sigma_P = \sqrt{0.5^2 50^2 + 2 \times 0.5 \times 0.5 \times 50 \times 50 \times (-0.5) + 0.5^2 50^2} = 25\%$$

$$\sigma_P < \sigma_A = \sigma_B$$

7-2 solution

In which of the cases is the volatility lower than that of the original stocks? (Select the best choice below.)

- ☐ A. In cases (d) and (e).
- ☐ B. In all of the cases.
- ☐ C. In cases (b), (c), (d) and (e).
- ☐ D. In none of the cases.

Scenario 5: $\rho = -1$

Calculation:

$$\sigma_P = \sqrt{0.5^2 50^2 + 2 \times 0.5 \times 0.5 \times 50 \times 50 \times (-1) + 0.5^2 50^2} = 0\%$$

$$\sigma_P < \sigma_A = \sigma_B$$

- Scenario 1: $\sigma_P = \sigma_A = \sigma_B$
- Scenario 2, 3, 4, 5: $\sigma_P < \sigma_A = \sigma_B$
- C (less volatility in cases (b), (c), (d) and (e))

7-4

A short-selling portfolio!

4. A hedge fund has created a portfolio using just two stocks. It has shorted \$20,000,000 worth of Oracle stock and has purchased \$86,000,000 of Intel stock. The correlation between Oracle's and Intel's returns is 0.65. The expected returns and standard deviations of the two stocks are given in the table below:

	Expected Return	Standard Deviation
Oracle	12.47%	46.06%
Intel	14.99%	38.85%

Suppose the correlation between Intel and Oracle's stock increases, but nothing else changes. Would the portfolio be more or less risky with this change?

(Select the best choice below.)

- ☐ A. More risky.
- ☐ B. Riskiness of the portfolio stays the same.
- ☐ C. Less risky.
- ☐ D. Cannot say without knowing how investors trade off expected return and volatility.

7-4 solution

Setup

- Short Oracle stock: -20m
- Purchase Intel stock: 86m
- Then our investment is -20m+86m=66m

Portfolio weight formula

$$k_i = \frac{\text{Value of Stock } i}{\text{Total Value of Portfolio}}$$

$$\bullet k_{oracle} = \frac{-20}{66} = -0.3030, k_{intel} = \frac{86}{66} = 1.3030$$

7-4 solution

Portfolio risk formula

$$\sigma_P = \sqrt{k^2 \sigma_A^2 + 2k(1 - k)\sigma_A \sigma_B \rho_{AB} + (1 - k)^2 \sigma_B^2}$$

- *Given that* $k = -0.3030 < 0$
- Then $2k(1 - w)\sigma_A \sigma_B < 0$
- Ceteris paribus, If $\rho_{AB}^{\uparrow} \rightarrow (2k(1 - k)\sigma_A \sigma_B \rho_{AB})^{\downarrow} \rightarrow \sigma_P^{\downarrow}$
- Thus, **C** is correct. (less risky)

7-3

3. A hedge fund has created a portfolio using just two stocks. It has shorted \$21,000,000 worth of Oracle stock and has purchased \$98,000,000 of Intel stock. The correlation between Oracle's and Intel's returns is 0.61. The expected returns and standard deviations of the two stocks are given in the table below:

$$\rho = 0.61$$

	Expected Return	Standard Deviation
Oracle	12.57%	44.57%
Intel	14.06%	40.94%

- a. What is the expected return of the hedge fund's portfolio?
b. What is the standard deviation of the hedge fund's portfolio?

- a. What is the expected return of the hedge fund's portfolio?

The expected return of the hedge fund's portfolio is _____. (Round to two decimal places.)

- b. What is the standard deviation of the hedge fund's portfolio?

The standard deviation of the hedge fund's portfolio _____. (Round to one decimal place.)

7-3 solution

Setup:

- Short Oracle stock: -21m
- Purchase Intel stock: 98m
- Then our investment is -21m+98m=77m

Portfolio weight formula:

$$k_i = \frac{\text{Value of Stock } i}{\text{Total Value of Portfolio}}$$

$$\bullet k_{oracle} = \frac{-21}{77} = -0.2727, k_{intel} = \frac{98}{77} = 1.2727$$

7-3 solution

	Expected Return	Standard Deviation
Oracle	12.57%	44.57%
Intel	14.06%	40.94%
	μ	σ

Portfolio return formula:

$$\mu_p = k\mu_A + (1 - k)\mu_B$$

- $\mu_p = (-0.2727) \times 12.57\% + (1.2727) \times 14.06\% = 14.47\%$

Portfolio risk formula:

$$\sigma_p = \sqrt{k^2\sigma_A^2 + 2k(1 - k)\sigma_A\sigma_B\rho_{AB} + (1 - k)^2\sigma_B^2}$$

$$\sigma_p = \sqrt{(-0.2727)^2 \times 44.57^2 + 2 \times (-0.2727) \times 1.2727 \times 44.57 \times 40.94 \times 0.61 + 1.2727^2 \times 40.94^2} = 45.72\%$$

Recap: Modern portfolio theory

- If we can collect the **historical** data and would like to know the average return and volatility.
- Then, we can take the entire portfolio as a single, standalone asset, and use the following formulas:
- Average return:

$$\overline{R_p} = \frac{1}{n} \sum_t R_{p,t}$$

- Volatility:

$$SD(p) = \sigma_p = \sqrt{\frac{\sum (R_{p,t} - \overline{R_p})^2}{n - 1}}$$

7-1

1. Using the data in the following table, calculate the volatility (standard deviation) of a portfolio that is 68% invested in stock A and 32% in stock B.

Year	2004	2005	2006	2007	2008	2009
Stock A	- 14%	15%	6%	- 8%	5%	5%
Stock B	13%	20%	1%	- 4%	- 9%	31%

The volatility of the portfolio is _____%. (Round to two decimal places.)

7-1 solution

Method 1: MPT approach

Core formulas:

$$\sigma_p = \sqrt{k^2 \sigma_A^2 + 2k(1-k) \sigma_A \sigma_B \rho_{AB} + (1-k)^2 \sigma_B^2}$$
$$\rho_{AB} = \frac{\sigma_{AB}}{\sigma_A \sigma_B} \rightarrow \sigma_A \sigma_B \rho_{AB} = \sigma_{AB}$$
$$\sigma_{AB} = \frac{1}{n-1} \sum_t (R_{A,t} - \bar{R}_A)(R_{B,t} - \bar{R}_B)$$
$$\bar{R}_A = \frac{1}{n} \sum_t R_{A,t} \quad \sigma_A = \sqrt{\frac{\sum (R_{A,t} - \bar{R}_A)^2}{n-1}}$$

7-1 solution

For Stock A

$$\overline{R}_A = \frac{1}{n} \sum_t R_{A,t}$$

Year	2004	2005	2006	2007	2008	2009
Stock A	- 14%	15%	6%	- 8%	5%	5%

$$\bullet \overline{R}_A = \frac{1}{6} \times (-14\% + 15\% + 6\% - 8\% + 5\% + 5\%) = 1.5\%$$

$$\sigma_A = \sqrt{\frac{\sum (R_{A,t} - \overline{R}_A)^2}{n - 1}}$$

$$\bullet \sigma_A = \sqrt{\frac{(-14\% - 1.5\%)^2 + (15\% - 1.5\%)^2 + \dots + (5\% - 1.5\%)^2}{6 - 1}} = 10.56\%$$

7-1 solution

For Stock B

$$\overline{R}_B = \frac{1}{n} \sum_t R_{B,t}$$

Stock B	13%	20%	1%	- 4%	- 9%	31%
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- $\overline{R}_B = \frac{1}{6} \times (13\% + 20\% + 1\% - 4\% - 9\% + 31\%) = 8.67\%$

$$\sigma_B = \sqrt{\frac{\sum (R_{B,t} - \overline{R}_B)^2}{n - 1}}$$

- $\sigma_B = \sqrt{\frac{(-13\% - 8.67\%)^2 + (20\% - 8.67\%)^2 + \dots + (31\% - 8.67\%)^2}{6 - 1}} = 15.34\%$

7-1 solution

Formula for the covariance between A and B

$$\sigma_{AB} = \frac{1}{n-1} \sum_t (R_{A,t} - \bar{R}_A)(R_{B,t} - \bar{R}_B)$$

Year	2004	2005	2006	2007	2008	2009
Stock A	-14%	15%	8%	-8%	5%	5%
Stock B	13%	20%	1%	-4%	-9%	31%

- $$\sigma_{AB} = \frac{1}{6-1} \times [(-14\% - 1.5\%) \times (13\% - 8.67\%) + (15\% - 1.5\%) \times (20\% - 8.67\%) + \dots + (5\% - 1.5\%) \times (31\% - 8.67\%)] = \frac{188}{5} = 37.6\%$$

7-1 solution

- Portfolio volatility formula:

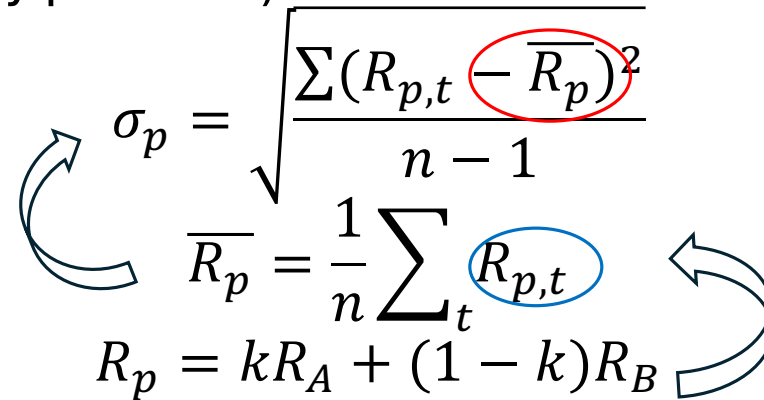
$$\sigma_p = \sqrt{k^2 \sigma_A^2 + 2k(1-k)\sigma_A\sigma_B\rho_{AB} + (1-k)^2 \sigma_B^2}$$

- $\sigma_p =$
 $\sqrt{0.68^2 \times 10.56\%^2 + 2 \times 0.68 \times 0.32 \times 37.6\% + 0.32^2 \times 15.34\%^2} =$
9.59%

7-1 solution

Method 2: Historical data approach

Core formulas (risky portfolio):

$$\sigma_p = \sqrt{\frac{\sum (R_{p,t} - \bar{R}_p)^2}{n - 1}}$$
$$\bar{R}_p = \frac{1}{n} \sum_t R_{p,t}$$
$$R_p = kR_A + (1 - k)R_B$$


7-1 solution

$$R_p = kR_A + (1 - k)R_B, \quad k = 0.68, 1 - k = 0.32$$

Year	Stock A	Stock B	Portfolio realized return (R_p)
2004	-14%	13%	$0.68(-14) + 0.32(13) = -5.36\%$
2005	15%	20%	$0.68(15) + 0.32(20) = 16.60\%$
2006	6%	1%	$0.68(6) + 0.32(1) = 4.40\%$
2007	-8%	-4%	$0.68(-8) + 0.32(-4) = -6.72\%$
2008	5%	-9%	$0.68(5) + 0.32(-9) = 0.52\%$
2009	5%	31%	$0.68(5) + 0.32(31) = 13.32\%$

7-1 solution

$$\overline{R_p} = \frac{1}{n} \sum_t R_{p,t}$$

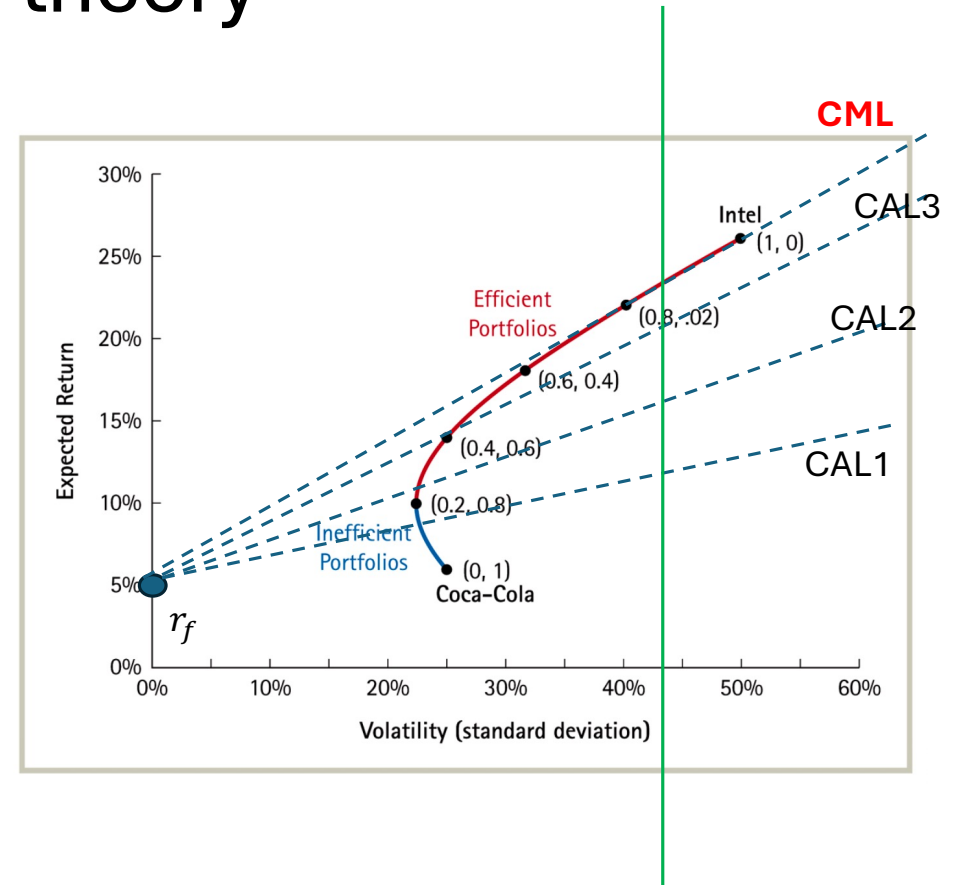
$$\bullet \overline{R_p} = \frac{-5.36+16.60+4.40-6.72+0.52+13.32}{6} = 3.79\%$$

$$\sigma_p = \sqrt{\frac{\sum (R_{p,t} - \overline{R_p})^2}{n - 1}}$$

$$\bullet \sigma_p = \sqrt{\frac{(-5.36-3.79)^2+(16.60-3.79)^2+\dots+(13.32-3.79)^2}{6-1}} = \sqrt{\frac{460.16}{5}} = 9.59\%$$

Recap: Modern portfolio theory

- Combining a **risk-free asset** with any risky portfolio creates a straight Capital Allocation Line (CAL).
- The line that is tangent to the efficient frontier is **Capital Market Line (CML)**.
- The CML represents the **optimal** complete portfolio, offering the highest possible Sharpe Ratio.



Recap: Modern portfolio theory

Core formulas (**complete portfolio, macro approach**):

- Portfolio risk formula:

$$\sigma_p = \sqrt{w^2 \sigma_{risky}^2 + 2w(1-w)\sigma_{risky}\sigma_f \rho_{riskyf} + (1-w)^2 \sigma_f^2}$$

- Given $\sigma_f = 0$ (the risk-free asset has zero volatility and zero correlation with everything)

$$\begin{aligned}\Rightarrow \sigma_p &= w\sigma_{risky} \\ \Rightarrow w &= \frac{\sigma_p}{\sigma_{risky}}\end{aligned}$$

Recap: Modern portfolio theory

- Expected return formula:

$$\mu_p = w\mu_{risky} + (1 - w)r_f$$

- Factor out the w :

$$\Rightarrow \mu_p = r_f + (\mu_{risky} - r_f)w$$

- Now, substitute the isolated w into this equation

$$\mu_p = r_f + (\mu_{risky} - r_f)\left(\frac{\sigma_p}{\sigma_{risky}}\right)$$

$$\Rightarrow \mu_p = r_f + \left(\frac{\mu_{risky} - r_f}{\sigma_{risky}}\right)\sigma_p$$

Sharpe ratio (constant)

Recap: Modern portfolio theory

Typical problem setup:

- You currently hold an **inefficient** portfolio.
- You are asked to “**upgrade**” to a more efficient complete portfolio by mixing a better risky asset (or the Tangency Portfolio) with the risk-free asset.

Two upgrade paths:

- a. Path a: Achieve a higher expected return while keeping the volatility fixed.
- b. Path b: Achieve a lower volatility while keeping the expected return fixed.

Recap: Modern portfolio theory

The four-step solution method

Step 1: Identify the target

- *Determine which parameter of your original portfolio you must match (either the target μ_p or the target σ_p).*

Step 2: Solve for the optimal weight (w)

- If matching risk: Set $\sigma_p = w\sigma_{risky} \Rightarrow$ solve for w .*
- If matching return: Set $\mu_p = w\mu_{risky} + (1 - w)r_f \Rightarrow$ solve for w .*

Recap: Modern portfolio theory

Step 3: Calculate dollar allocations

- *Multiply your optimal weights (w and $w - 1$) by the total investment capital to get exact dollar amounts.*

Step 4: Verify the Upgrade

- *Calculate the newly improved parameter (the maximized μ_p or the minimized σ_p) to ensure the portfolio is actually better.*

7-5

-
5. You currently have \$80,000 invested in a portfolio that has an expected return of 10% and a volatility of 9%. Suppose the risk-free rate is 4%, and there is another portfolio has an expected return of 18% and a volatility of 11%.
- a. What portfolio has a higher expected return than your portfolio but with the same volatility?
 - b. What portfolio has a lower volatility than your portfolio but with the same expected return?
-

a. What portfolio has a higher expected return than your portfolio but with the same volatility? (Round all intermediate values to five decimal places as needed.)

The portfolio should be composed of \$ _____ in the other portfolio, and the remaining \$ _____ in the risk-free investment. (Round to the nearest dollar.)

b. What portfolio has a lower volatility than your portfolio but with the same expected return? (Round all intermediate values to five decimal places as needed.)

The portfolio should be composed of \$ _____ in the other portfolio, and the remaining \$ _____ in the risk-free investment. (Round to the nearest dollar.)

7-5 solution

Setup:

- Net investment: $V = \$80,000$
- Risk-free rate: $r_f = 4\%$
- Current portfolio (the benchmark):
 - Expected return $\mu_{current} = 10\%$; Volatility $\sigma_{current} = 9\%$
- “Another portfolio” (risky portfolio):
 - Expected return $\mu_{risky} = 18\%$; Volatility $\sigma_{risky} = 11\%$

7-5 solution

a.

Step 1: Identify the target

a. What portfolio has a higher expected return than your portfolio but with the same volatility?

- **Target:** Achieve a higher μ_p while keeping $\sigma_p = 9\%$

Step 2: Solve for the optimal weight (w)

- *If matching risk: Set $\sigma_p = w\sigma_{risky} \Rightarrow$ solve for w .*

$$\begin{aligned}\sigma_p &= w\sigma_{risky} \\ \Rightarrow w &= \frac{\sigma_p}{\sigma_{risky}}\end{aligned}$$

7-5 solution

- *Given our target $\sigma_p = 9\%$, and new portfolio's $\sigma_{risky} = 11\%$:*
- *Then $w = \frac{0.09}{0.11} = 0.8182$*

Step 3: Calculate dollar allocations

- *So the dollar value of the other investment is $0.8182 \times \$80,000 = \$65,454.55$. And the remaining $\$14,545.45$ in a risk-free asset.*

7-5 solution

Step 4: Verify the Upgrade

Expected return formula:

$$\mu_p = w\mu_{risky} + (1 - w)r_f$$

- $\mu_p = 81.82\% \times 18\% + 18.18\% \times 4\% = 15.45\% > 10\%$
- Conclusion: The new portfolio achieves higher return than the original one, while maintaining the same volatility (9%).

7-5 solution

b.

Step 1: Identify the target

b. What portfolio has a lower volatility than your portfolio but with the same expected return?

Target: Achieve a lower σ_p while keeping $\mu_p = 10\%$

Step 2: Solve for the optimal weight (w)

- *If matching return: Set $\mu_p = w\mu_{risky} + (1 - w)r_f \Rightarrow$ solve for w .*
- $\mu_p = w\mu_{risky} + (1 - w)r_f = 10\%$
- $w \times 18\% + (1 - w) \times 4\% = 10\% \Rightarrow w = 0.4286$

7-5

Step 3: Calculate dollar allocations

- So the dollar value of the other investment is $0.4286 \times \$80,000 = \$34,285.71$. And the remaining \$45,714.29 in a risk-free asset.

Step 4: Verify the Upgrade

Verify the new risk:

$$\sigma_p = w\sigma_{risky}$$

- $\sigma_p = 0.4286 \times 11\% = 4.72\% < 9\%$
- Conclusion: The new portfolio minimizes risk while matching the same target return (10%).

7-6

6. Your investment portfolio consists of \$16,000 invested in only one stock—Microsoft. Suppose the risk-free rate is 5%, Microsoft stock has an expected return of 12% and a volatility of 43%, and the market portfolio has an expected return of 10% and a volatility of 17%. *Hint: Make sure to round all intermediate calculations to at least five decimal places.*

Under the CAPM assumptions,

- What alternative investment has the lowest possible volatility while having the same expected return as Microsoft? What is the volatility of this investment?
- What investment has the highest possible expected return while having the same volatility as Microsoft? What is the expected return of this investment?

a. What alternative investment has the lowest possible volatility while having the same expected return as Microsoft?

To create an alternative investment that has the lowest possible volatility while having the same expected return as Microsoft, we use the following strategy:

Sell: \$ _____ worth of Microsoft stock. (Round to the nearest dollar.)

Borrow: \$ _____ at the risk-free rate. (Round to the nearest dollar.)

Buy: \$ _____ worth of the market portfolio. (Round to the nearest dollar.)

Buy: \$ _____ worth of the risk-free investment. (Round to the nearest dollar.)

This portfolio has a volatility of _____ %. (Round to two decimal places.)

7-6 solution

Setup:

- Net investment: $V = \$16,000$
- Risk-free rate: $r_f = 5\%$
- Microsoft: $\mu_{MSFT} = 12\%, \sigma_{MSFT} = 43\%$
- Market portfolio: $\mu_T = 10\%, \sigma_T = 17\%$

7-6 solution

Step 1: Identify the target

a. What alternative investment has the lowest possible volatility while having the same expected return as Microsoft?

Target: Same return ($\mu_p = 12\%$), lowest volatility

Step 2: Solve for the optimal weight (w)

- *If matching return: Set $\mu_p = w\mu_T + (1 - w)r_f \Rightarrow$ solve for w .*
- $12\% = w(10\%) + (1 - w)(5\%)$
- $7\% = 5\%w \Rightarrow w = 1.4$

7-6 solution

Sell: \$ _____ worth of Microsoft stock. (Round to the nearest dollar.)
Borrow: \$ _____ at the risk-free rate. (Round to the nearest dollar.)
Buy: \$ _____ worth of the market portfolio. (Round to the nearest dollar.)
Buy: \$ _____ worth of the risk-free investment. (Round to the nearest dollar.)

Step 3: Calculate dollar allocations

- **Sell:** \$16,000 worth of Microsoft stock (exit the inefficient position)
- **Borrow:** Since $w = 1.4$, we need to invest 140% of our capital. We must borrow 40% of our \$16,000.
 - $0.4 \times \$16,000 = \$6,400$ *at the risk – free rate*
- **Buy:** \$22,400 worth of the market portfolio ($1.4 \times \$16,000$).
- **Buy:** \$0 worth of the risk-free investment (Because we are borrower, not a lender).

7-6 solution

This portfolio has a volatility of _____%. (Round to two decimal places.)

Step 4: Verify the Upgrade

$$\sigma_p = w\sigma_T$$

- $\sigma_p = 1.4 \times 17\% = 23.80\%$

Verification:

- $\sigma_p = 23.80\% < \sigma_{MSFT} = 43\%$

Conclusion:

At the same expected return level (12%), this strategy reduces volatility by **19.20%** (from 43% to 23.80%) compared to the original Microsoft-only portfolio.

7-6 (cont.)

b. What investment has the highest possible expected return while having the same volatility as Microsoft?

To create an alternative investment that has the highest possible expected return while having the same volatility as Microsoft, we use the following strategy:

Sell: \$ _____ worth of Microsoft stock. (Round to the nearest dollar.)

Borrow: \$ _____ at the risk-free rate. (Round to the nearest dollar.)

Buy: \$ _____ worth of the market portfolio. (Round to the nearest dollar.)

Buy: \$ _____ worth of the risk-free investment. (Round to the nearest dollar.)

The expected return of this investment will be _____%. (Round to two decimal places.)

7-6 solution (cont.)

Step 1: Identify the target

b. What investment has the highest possible expected return while having the same volatility as Microsoft?

Target: Same volatility ($\sigma_T = 43\%$), highest return

Step 2: Solve for the optimal weight (w)

- *If matching risk: Set $\sigma_p = w\sigma_{risky} \Rightarrow$ solve for w .*

$$\sigma_p = w\sigma_T$$

- $43\% = w \times 17\%$

- $\Rightarrow w = \frac{43}{17} = 2.52941$

7-6 solution (cont.)

Step 3: Calculate dollar allocations

- **Sell:** \$16,000 worth of Microsoft stock.
- **Borrow:** We need to borrow $w - 1$ of our initial capital.
 - $(2.52941 - 1) \times \$16,000 = \$24,471$ *at the risk-free rate*
- **Buy:** $2.52941 \times 16,000 = \$40,471$ worth of the market portfolio.
- **Buy:** \$0 worth of the risk-free investment.

7-6 solution (cont.)

Step 4: Verify the Upgrade

$$\mu_p = w\mu_T + (1 - w)r_f$$

- $\mu_p = 2.52941(10\%) + (1 - 2.52941)(5\%) = 17.65\%$

Verification:

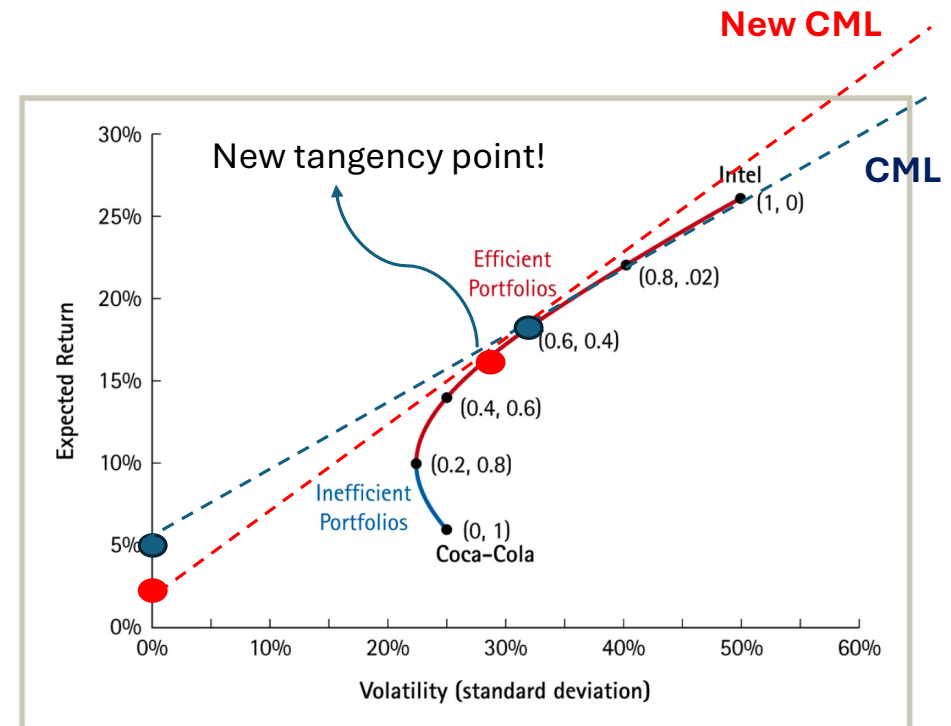
- $\mu_p = 17.65\% > \mu_{MSFT} = 12\%$

Conclusion:

At the same volatility level (43%), this strategy generates a **5.65%** higher expected return compared to the original Microsoft-only portfolio.

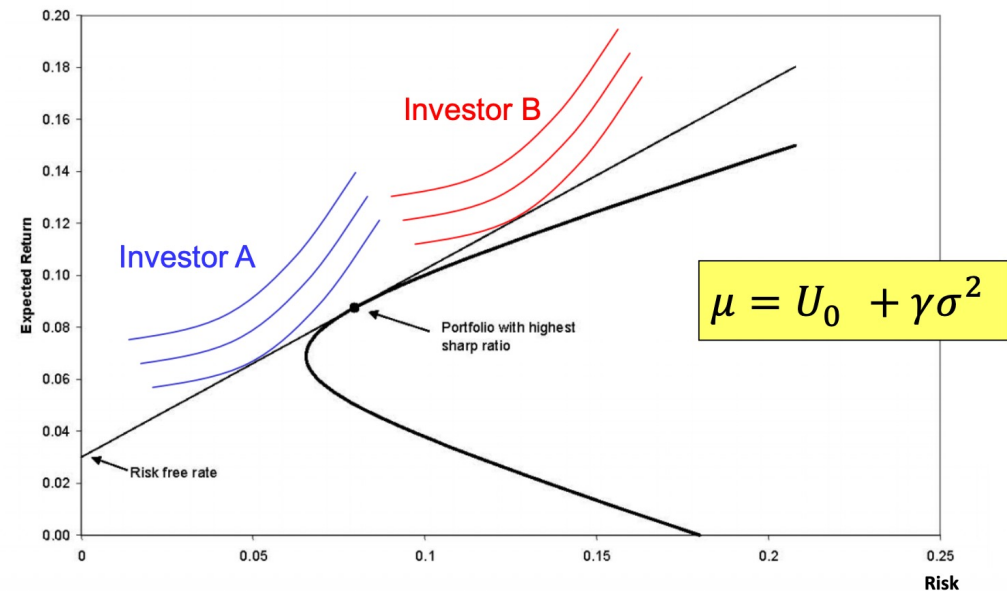
Recap: Mutual fund separation theorem

- So far, we have assumed that both the composition of the Tangency Portfolio and the allocation weights are simply given to us.
- However, with changes in the risk-free rate, the original Tangency Portfolio is **no longer efficient**. We need the ability to find a new one.



Recap: Mutual fund separation theorem

- Furthermore, investors have **varying risk appetites**. Although rational investors should hold the exact same Tangency Portfolio, their allocation proportions between the risky portfolio and the risk-free asset are **totally different**.



Recap: Mutual fund separation theorem

- The origin: Built upon Harry Markowitz's Nobel-winning Modern Portfolio Theory (MPT) and expanded by **James Tobin**.
- The big idea: All rational investment decisions can be split into two completely separate steps.
- You can separate the “Math” (objective, Step 1) from the “Feelings” (subjective, Step 2).



James Tobin (1918-2002)

AMERICANO VS ESPRESSO

4 OZ
HOT WATER

2 OZ
ESPRESSO



AMERICANO

1 OZ
ESPRESSO
SHOT



ESPRESSO



Recap: Mutual fund separation theorem

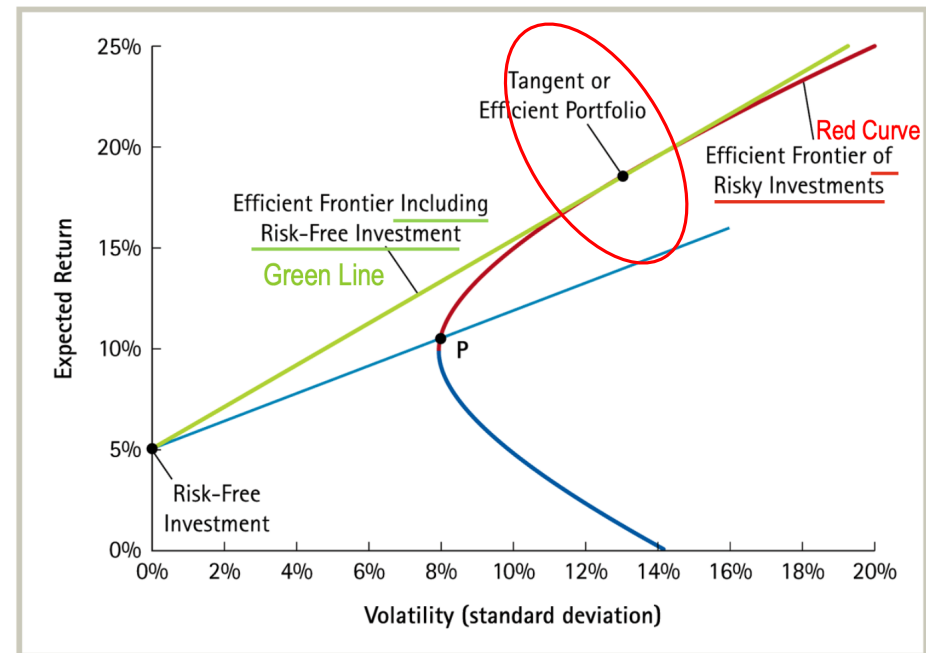
Step 1 – the “Math” (find the tangency portfolio)

- What it is: The single best mathematical combination of risky assets that maximizes the reward-to-risk ratio (Sharpe Ratio).
- The key takeaway: This optimal “Tangency Portfolio” is exactly the **same for everyone**.
- Application: You do not pick individual stocks based on your personality. You let the math find the optimal mix.

Recap: Mutual fund separation theorem

Step 1 – the “Math” (cont.)

- Key question: How to mathematically locate the exact Tangency Portfolio?
- Strategy: (1) Introduce the risk-free rate (r_f) and maximize the Sharpe Ratio ($\frac{\mu_p - r_f}{\sigma_p}$) to find the optimal tangent point.



Recap: Mutual fund separation theorem

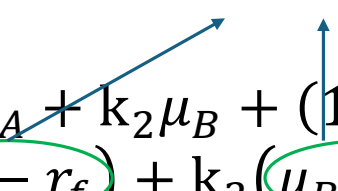
- *Note: The following derivation is non—examinable!*
- Setup: To simplify the math, we assume a universe with **only** two risky assets and one risk-free asset.
- Core formulas (**complete portfolio, micro approach**):
$$\mu_p = k_1\mu_A + k_2\mu_B + (1 - k_1 - k_2)r_f$$
$$\sigma_p^2 = k_1^2\sigma_A^2 + 2k_1k_2\sigma_A\sigma_B\rho_{AB} + k_2^2\sigma_B^2$$
- The approach: (1) **fix** the return of the portfolio (μ_p); (2) minimize the variance (σ_p^2) to find the optimal weights (k^T).

Recap: Mutual fund separation theorem

- We can write:

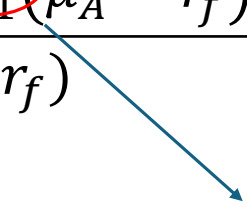
$$\begin{aligned}\mu_p &= k_1\mu_A + k_2\mu_B + (1 - k_1 - k_2)r_f \\ &= k_1(\mu_A - r_f) + k_2(\mu_B - r_f) + r_f\end{aligned}$$

Excess return



The diagram shows two blue arrows pointing upwards from the terms $(\mu_A - r_f)$ and $(\mu_B - r_f)$ in the second equation to the red text 'Excess return' located above the space between the two equations.

- Since the portfolio return (μ_p) is fixed, we can determine k_2 :

$$k_2 = \frac{(\mu_p - r_f) - k_1(\mu_A - r_f)}{(\mu_B - r_f)}$$


Only free variable

The diagram shows a blue arrow pointing downwards from the term $k_1(\mu_A - r_f)$ in the numerator of the equation to the red text 'Only free variable' located below the equation.

Recap: Mutual fund separation theorem

- Substitute the expression for k_2 into the variance formula:

$$\sigma_p^2 = k_1^2 \sigma_A^2 + 2k_1 k_2 \sigma_A \sigma_B \rho_{AB} + k_2^2 \sigma_B^2$$



$$\begin{aligned} \sigma_p^2 &= k_1^2 \sigma_A^2 + 2k_1 \frac{(\mu_p - r_f) - k_1(\mu_A - r_f)}{(\mu_B - r_f)} \sigma_A \sigma_B \rho_{AB} \\ &\quad + \frac{(\mu_p - r_f) - k_1(\mu_A - r_f)}{(\mu_B - r_f)}^2 \sigma_B^2 \end{aligned}$$

Recap: Mutual fund separation theorem

- Collect terms for k_1^2 and k_1

$$\begin{aligned} \sigma_p^2 &= k_1^2 \left(\sigma_A^2 - 2\sigma_A\sigma_B\rho_{AB} \frac{\mu_A - r_f}{\mu_B - r_f} + \sigma_B^2 \frac{(\mu_A - r_f)^2}{(\mu_B - r_f)^2} \right) \\ &+ k_1 \left(2\sigma_A\sigma_B\rho_{AB} \frac{\mu_p - r_f}{\mu_B - r_f} - 2\sigma_B^2 \frac{(\mu_p - r_f)(\mu_A - r_f)}{(\mu_B - r_f)^2} \right) + \sigma_B^2 \frac{(\mu_p - r_f)^2}{(\mu_B - r_f)^2} \end{aligned}$$

- To simplify, define the coefficients a, b and c:

$$\begin{aligned} a &= \sigma_A^2 - 2\sigma_A\sigma_B\rho_{AB} \frac{\mu_A - r_f}{\mu_B - r_f} + \sigma_B^2 \frac{(\mu_A - r_f)^2}{(\mu_B - r_f)^2} \\ b &= 2\sigma_A\sigma_B\rho_{AB} \frac{\mu_p - r_f}{\mu_B - r_f} - 2\sigma_B^2 \frac{(\mu_p - r_f)(\mu_A - r_f)}{(\mu_B - r_f)^2} \\ c &= \sigma_B^2 \frac{(\mu_p - r_f)^2}{(\mu_B - r_f)^2} \end{aligned}$$

Recap: Mutual fund separation theorem

- The variance simplifies to a quadratic equation:

$$\sigma_p^2 = ak_1^2 + bk_1 + c$$

- Apply the **First-Order Condition (FOC)** to minimize variance:

$$\begin{aligned}(\sigma_p^2(k_1))' &= 2ak_1^* + b = 0 \\ \Rightarrow k_1^* &= \frac{-b}{2a}\end{aligned}$$

- Substituting back the original expression for a and b:

$$\Rightarrow k_1^T = \frac{k_1^*}{k_1^* + k_2^*} = \frac{E_1\sigma_2^2 - E_2\rho\sigma_1\sigma_2}{E_2\sigma_1^2 - (E_1 + E_2)\rho\sigma_1\sigma_2 + E_1\sigma_2^2}$$

Recap: Mutual fund separation theorem

$$k_2^T = \frac{k_2^*}{k_1^* + k_2^*} = \frac{E_2\sigma_1^2 - E_1\rho\sigma_1\sigma_2}{E_2\sigma_1^2 - (E_1 + E_2)\rho\sigma_1\sigma_2 + E_1\sigma_2^2}$$

⇒

$$\text{Or simply: } k_2^T = 1 - k_1^T$$

where $E_1 = \mu_1 - r_f$, $E_2 = \mu_2 - r_f$ are the excess returns of asset 1 and asset 2, respectively

Recap: Mutual fund separation theorem

- Finalizing Step 1: By applying the optimal tangency weights (k_1^T , k_2^T), we can compute the exactly expected return (μ_T) and volatility (σ_T) of the Tangency Portfolio:

$$\mu_T = k_1^T \mu_A + (1 - k_1^T) \mu_B$$
$$\sigma_T = \sqrt{(k_1^T)^2 \sigma_A^2 + 2k_1^T(1 - k_1^T)\sigma_A\sigma_B\rho_{AB} + (1 - k_1^T)^2 \sigma_B^2}$$

Recap: Mutual fund separation theorem

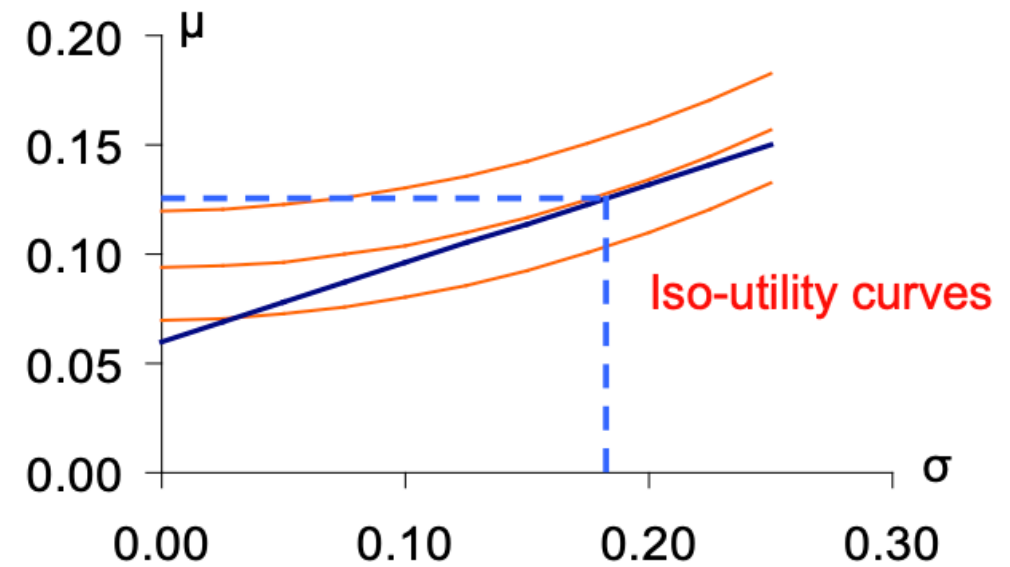
Step 2 – the “Feelings” (allocate your capital)

- What it is: Deciding how to split your total money between the Tangency Portfolio and a Risk-Free Asset (Cash / T-bills).
- The key takeaway: This step is unique to you and depends entirely on your personal risk tolerance.
- The choices:
 - Conservative: 80% cash + 20% tangency portfolio
 - Moderate: 50% cash + 50% tangency portfolio
 - Aggressive: 100% tangency portfolio

Recap: Mutual fund separation theorem

Step 2 – the “Feelings” (cont.)

- Key question: How to mathematically determine an individual’s optimal capital allocation (w)?
- Strategy: (1) Introduce the utility function to quantify person risk aversion (γ), and (2) maximize the utility using the FOC to find the optimal weight.



Recap: Mutual fund separation theorem

- The **mean-variance utility function** describes investors who only care about
 - means, which they like, and
 - variances, which they dislike.

$$U = \mu - \gamma \sigma^2$$

- where γ is the investor's risk-aversion coefficient

Recap: Mutual fund separation theorem

- Substituting the expressions for return and volatility into:

$$U = \mu - \gamma \sigma^2$$
$$\Rightarrow U(w) = \left((\mu_T - r_f)w + r_f \right) - \gamma (w^2 \sigma_T^2)$$

- Using the First-Order Condition (FOC) to maximize utility,

$$U(w)' = \mu_T - r_f - 2\gamma w \sigma_T^2 = 0$$

- So that

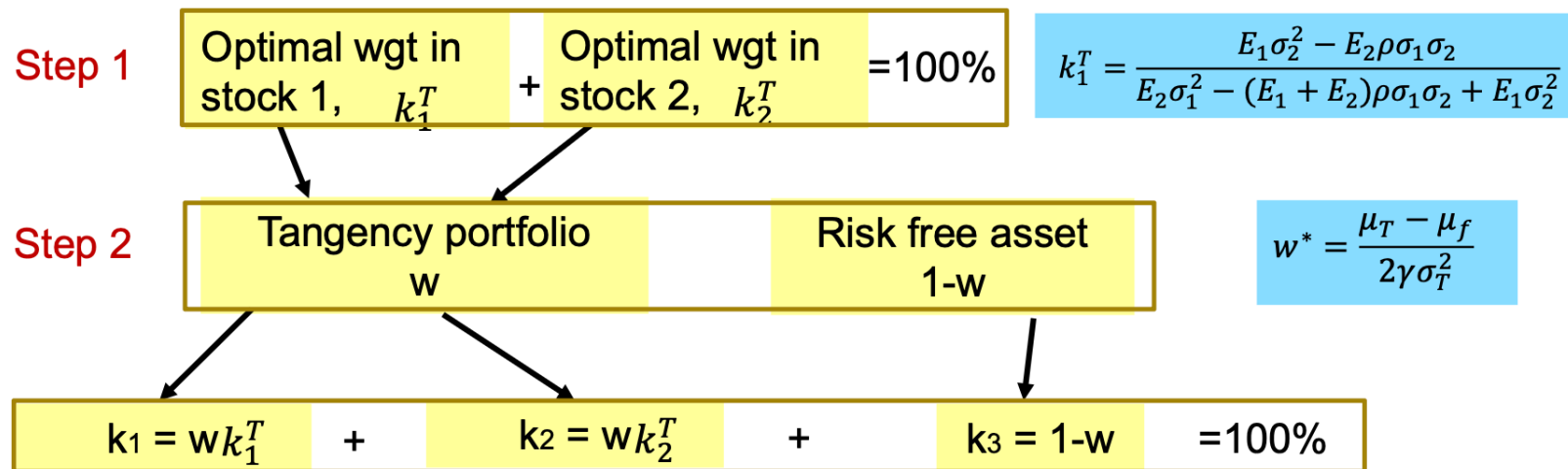
$$w^* = \frac{\mu_T - \mu_f}{2\gamma \sigma_T^2}$$

Recap: Mutual fund separation theorem

- For investors, they can allocate $w^* k_1^T$ in Asset A, $w^* k_2^T$ in Asset B, and $(1 - w^*)$ in the risk-free asset to build their own optimal investment portfolio (k_1, k_2, k_3) .
- Now, the optimal return and risk for a specific investor can be determined by substituting w^* back into the complete portfolio formulas:

$$\mu_p = (\mu_T - r_f) \left(\frac{\mu_T - r_f}{2\gamma\sigma_T^2} \right) + r_f$$
$$\sigma_p = \left(\frac{\mu_T - r_f}{2\gamma\sigma_T^2} \right) \sigma_T = \frac{\mu_T - r_f}{2\gamma\sigma_T}$$

Recap: Mutual fund separation theorem



7-8

8. The following symbols are used for this portfolio optimization question.

Weight in the optimal risky portfolio: w

Weight in the risk-free rate: $1-w = k_3$

Fraction of investment in risky asset 1: k_1

Fraction of investment in risky asset 2: k_2

so that $k_1 + k_2 = w$.

For the portfolio consisting of the stocks 1 and 2 and the riskfree asset with the following characteristics (all expressed in annual terms) :

$$\mu_1 = 6\%, \quad \mu_2 = 8\%, \quad \sigma_1 = 10\%, \quad \sigma_2 = 15\%, \quad \rho_{12} = -.2, \quad r_f = 3\%$$

- (1) Plot the risk-return diagram.
- (2) Calculate expected return and standard deviation of the tangency portfolio, where the investment strategy is $k_T = (k_1^T, k_2^T)$.

7-8 solution

a.

Core formulas (**risky portfolio**):

$$\mu_p = k\mu_A + (1 - k)\mu_B$$
$$\sigma_p = \sqrt{k^2\sigma_A^2 + 2k(1 - k)\sigma_A\sigma_B\rho_{AB} + (1 - k)^2\sigma_B^2}$$

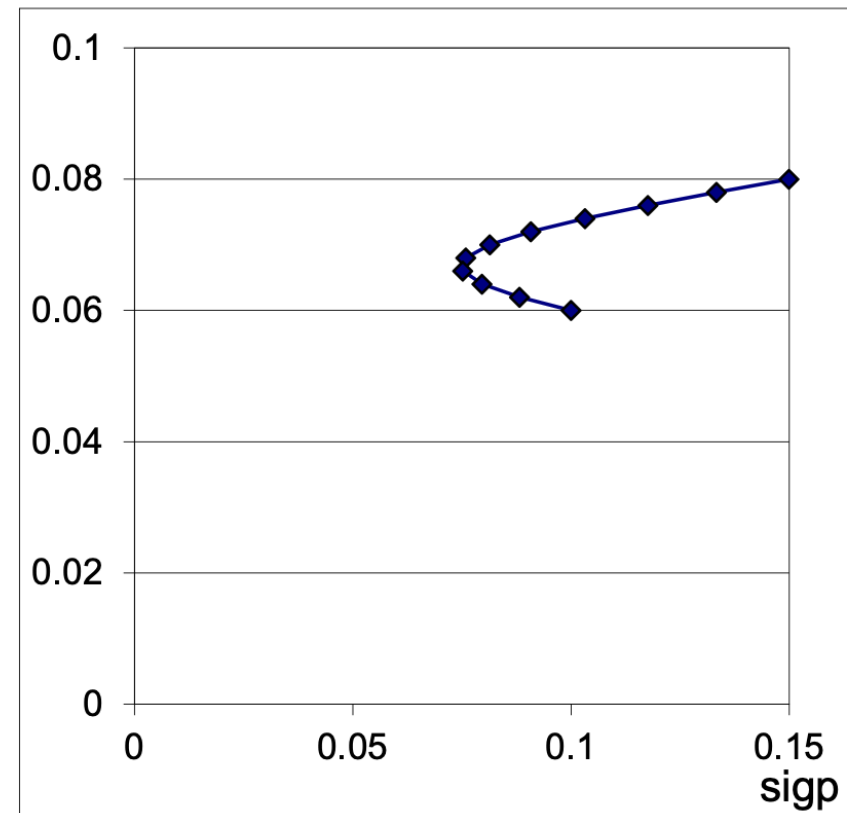
Setup:

$$\mu_1 = 6\%, \quad \mu_2 = 8\%, \quad \sigma_1 = 10\%, \quad \sigma_2 = 15\%, \quad \rho_{12} = -.2, \quad r_f = 3\%$$

- $\Rightarrow$ The only unknown variable is the portfolio fractions (k).

7-8 solution

k	1-k	$E(r_p)$	$\sigma(r_p)$
0	1	0.08	0.15
0.1	0.9	0.078	0.13336
0.2	0.8	0.076	0.117644
0.3	0.7	0.074	0.103271
0.4	0.6	0.072	0.090885
0.5	0.5	0.07	0.081394
0.6	0.4	0.068	0.075895
0.7	0.3	0.066	0.075266
0.8	0.2	0.064	0.079624
0.9	0.1	0.062	0.088233
1	0	0.06	0.1



7-8 solution

b.

$$k_1^T = \frac{k_1^*}{k_1^* + k_2^*} = \frac{E_1\sigma_2^2 - E_2\rho\sigma_1\sigma_2}{E_2\sigma_1^2 - (E_1 + E_2)\rho\sigma_1\sigma_2 + E_1\sigma_2^2}$$

$$k_2^T = \frac{k_2^*}{k_1^* + k_2^*} = \frac{E_2\sigma_1^2 - E_1\rho\sigma_1\sigma_2}{E_2\sigma_1^2 - (E_1 + E_2)\rho\sigma_1\sigma_2 + E_1\sigma_2^2}$$

Or simply: $k_2^T = 1 - k_1^T$

where $E_1 = \mu_1 - r_f$, $E_2 = \mu_2 - r_f$ are the excess returns of asset 1 and asset 2, respectively

7-8 solution

Calculation:

$$\mu_1 = 6\%, \quad \mu_2 = 8\%, \quad \sigma_1 = 10\%, \quad \sigma_2 = 15\%, \quad \rho_{12} = -.2, \quad r_f = 3\%$$

- The excess returns are $E_1 = \mu_1 - r_f = 6\% - 3\% = 3\%$ and $E_2 = \mu_2 - r_f = 8\% - 3\% = 5\%$.

$$k_1^T = \frac{k_1^*}{k_1^* + k_2^*} = \frac{E_1\sigma_2^2 - E_2\rho\sigma_1\sigma_2}{E_2\sigma_1^2 - (E_1 + E_2)\rho\sigma_1\sigma_2 + E_1\sigma_2^2}$$

7-8 solution

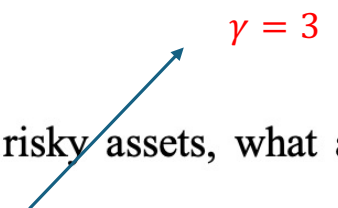
- $k_1^T = \frac{0.03(0.15)^2 - 0.05(-0.2)(0.1)(0.15)}{0.05(0.1)^2 - (0.03 + 0.05)(-0.2)(0.1)(0.15) + 0.03(0.15)^2} = 0.58304$
- $k_2^T = 1 - k_1^T = 1 - 0.5830 = 0.41696$

Core formulas (tangency portfolio):

$$\mu_T = k_1^T \mu_A + (1 - k_1^T) \mu_B$$
$$\sigma_T = \sqrt{(k_1^T)^2 \sigma_A^2 + 2k_1^T(1 - k_1^T) \sigma_A \sigma_B \rho_{AB} + (1 - k_1^T)^2 \sigma_B^2}$$

- $\mu_T = 0.58304(0.06) + 0.41696(0.08) = 0.06834$
- $\sigma_T = 0.0765$

7-8 (cont.)

- (3) If we invest 20% in the risk-free asset and 80% in the two risky assets, what are the fractions of capital to be invested in stock 1 and 2.
- (4) The utility of an investment strategy is given by $U = \mu(w) - 3\sigma^2(w)$, where $\mu(w)$ is the expected return and $\sigma^2(w)$ is the variance of the strategy $k=(k_1, k_2, k_3)$. Here k_1 is the proportion in stock 1, k_2 is the proportion in stock 2 and k_3 is the proportion in the risk-free asset. Use the results from part (2) above and the mutual fund separation theorem to calculate the strategy k^* that maximizes the given utility.
- 
- $\gamma = 3$

7-8 solution (cont.)

c.

- Weight in stock 1: $k_1 = wk_1^T = 0.8(0.58304) = 0.4664$
- Weight in stock 2: $k_2 = wk_2^T = 0.8(0.41696) = 0.3336$

d.

$$w^* = \frac{\mu_T - \mu_f}{2\gamma\sigma_T^2}$$

- $w^* = \frac{0.0634 - 0.03}{2(3)(0.0765)^2} = 1.0908$

7-8 solution (cont.)

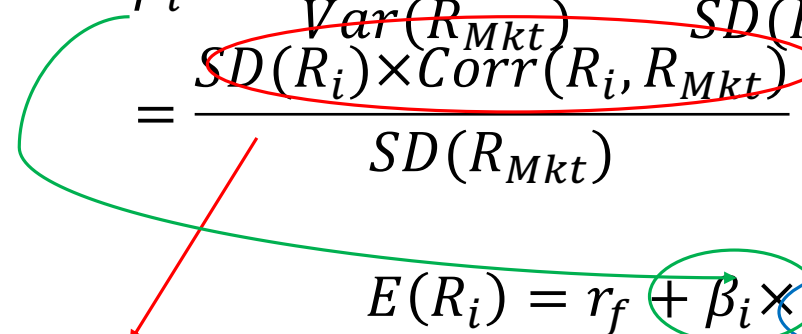
- As the allocation between the two stocks is fixed to be $(k_1^T, k_2^T) = (0.58304, 0.41696)$, the final fraction invested in the two stocks are:
- $(k_1, k_2) = (w^* k_1^T, w^* k_2^T) = (1.0908 \times 0.58304, 1.0908 \times 0.41696) = (0.636, 0.455)$
- Therefore, the optimal investment strategy is $k = (k_1, k_2, k_3) = (0.636, 0.455, 1 - 0.636 - 0.455) = (0.636, 0.455, -0.091)$

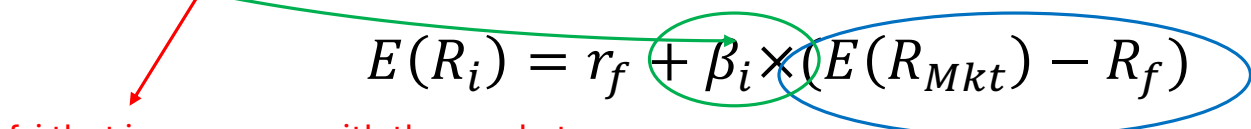
Recap: CAPM

- Previously, we focused on the total risk (systematic + idiosyncratic) of the entire portfolio. We simply took the expected returns of individual assets as given constants.
- But where do individual asset returns come from? Since rational investors hold the fully diversified Market Portfolio, all idiosyncratic risk is **diversified away**.
- Therefore, the market **only rewards** investors for bearing systematic risk. The Capital Asset Pricing Model (CAPM) allows us to calculate the exact required return of any asset based purely on its systematic risk (β).

Recap: CAPM

Core formulas (CAPM):

$$\beta_i = \frac{Cov(R_i, R_{Mkt})}{Var(R_{Mkt})} = \frac{SD(R_i)}{SD(R_{Mkt})} \times \frac{Cov(R_i, R_{Mkt})}{SD(R_i)SD(R_{Mkt})}$$
$$= \frac{SD(R_i) \times Corr(R_i, R_{Mkt})}{SD(R_{Mkt})}$$


$$E(R_i) = r_f + \beta_i \times (E(R_{Mkt}) - R_f)$$


Volatility of i that is common with the market

Risk premium for security i

7-7

7. Consider a portfolio consisting of the following three stocks:

	Portfolio Weight	Volatility	Correlation with the Market Portfolio
HEC Corp	0.24	10%	0.41
Green Midget	0.35	28%	0.65
AliveAndWell	0.41	12%	0.53

The volatility of the market portfolio is 10% and it has an expected return of 8%. The risk-free rate is 3%.

- Compute the beta and expected return of each stock.
- Using your answer from part (a), calculate the expected return of the portfolio.
- What is the beta of the portfolio?
- Using your answer from part (c), calculate the expected return of the portfolio and verify that it matches your answer to part (b).

- Compute the beta and expected return of each stock.

(Round to two decimal places.)

	Portfolio Weight (A)	Volatility (B)	Correlation (C)	Beta (D)	Expected Return (E)
HEC Corp	0.24	10%	0.41		%
Green Midget	0.35	28%	0.65		%
AliveAndWell	0.41	12%	0.53		%

7-7 solution

Setup:

- Risk-free rate (R_f): 3%
- Market expected return $E(R_{Mkt})$: 8%
- Market risk premium: $E(R_{Mkt}) - R_f = 8\% - 3\% = 5\%$
- Market volatility (σ_{Mkt}): 10%

7-7 solution

$$\beta_i = \frac{SD(R_i) \times \text{Corr}(R_i, R_{Mkt})}{SD(R_{Mkt})}$$
$$E(R_i) = r_f + \beta_i \times (E(R_{Mkt}) - R_f)$$

Calculation:

For HEC Corp:

	Portfolio Weight	Volatility	Correlation with the Market Portfolio
HEC Corp	0.24	10%	0.41

- $\beta = 0.41 \times \frac{10\%}{10\%} = 0.41$
- $E(R_{HEC}) = 3\% + (0.41 \times 5\%) = 5.05\%$

7-7 solution

For Green Midget:

	w	σ	ρ
Green Midget	0.35	28%	0.65

$$\beta_i = \frac{SD(R_i) \times Corr(R_i, R_{Mkt})}{SD(R_{Mkt})}$$

- $\beta = 0.65 \times \frac{28\%}{10\%} = 1.82$

$$E(R_i) = r_f + \beta_i \times (E(R_{Mkt}) - R_f)$$

- $E(R_{GreenMidget}) = 3\% + (1.82 \times 5\%) = 12.10\%$

7-7 solution

For AliveAndWell:

	w	σ	ρ
AliveAndWell	0.41	12%	0.53

$$\beta_i = \frac{SD(R_i) \times Corr(R_i, R_{Mkt})}{SD(R_{Mkt})}$$

- $\beta = 0.53 \times \frac{12\%}{10\%} = 0.636$

$$E(R_i) = r_f + \beta_i \times (E(R_{Mkt}) - R_f)$$

- $E(R_{AliveAndWell}) = 3\% + (0.636 \times 5\%) = 6.18\%$

7-7 solution

b, c.

Core formulas:

$$E(R_p) = \sum (w_i \times E(R_i))$$

$$\beta_p = \sum (w_i \times \beta_i)$$

- $\mu_p = 0.24 \times 5.05\% + 0.35 \times 12.10\% + 0.41 \times 6.18\% = 7.98\%$
- $\beta_p = 0.24 \times 0.41 + 0.35 \times 1.82 + 0.41 \times 0.636 = 0.997$

7-7 solution

d.

$$E(R_p) = r_f + \beta_p \times (E(R_{Mkt}) - R_f)$$

- $E(R_p) = 3\% + 0.997 \times 5\% = 7.98\%$

Conclusion: the final result is 7.98%, which perfectly matches the calculation from part (b).